TinyGRPO-Math: Rule-Based GRPO Post-Training on GSM8K

Abstract

This report evaluates a compact GRPO post-training experiment for mathematical reasoning on GSM8K. A non-math-specialized base model, `Qwen/Qwen2.5-1.5B-Instruct`, was trained with deterministic rule-based rewards instead of a learned reward model. The reward combines exact final-answer correctness, final-line answer formatting, parseability, and penalties for invalid or overlong outputs. On 200 held-out GSM8K test examples, the 1000-update continuation model improved accuracy from **66.5%** to **71.0%** and final-answer format compliance from **77.5%** to **97.5%**, while preserving 100% parseability and zero truncation.

1 Experiment Summary

Goal. Adapt a from-scratch GRPO training loop from preference-style chatbot alignment to verifiable math reasoning. GSM8K is a good fit because final answers are objective and can be scored without training a reward model.

Model. The policy and KL reference use `Qwen/Qwen2.5-1.5B-Instruct`. I intentionally did not use a math-specialized checkpoint as the training target, since an already math-post-trained model would leave less room for visible improvement.

Data. Training used GSM8K train rows 0–999 in two stages. Evaluation used GSM8K test rows 0–199. An exact question-text overlap check between these 1000 train questions and 200 test questions found 0 overlaps.

Training. Each GRPO update sampled $G = 4$ completions for one prompt. The first stage trained on rows 0–499 for 500 updates. The continuation stage initialized the policy from the first stage checkpoint, kept the KL reference anchored to the original base model, and trained on rows 500–999 for another 500 updates. The continuation stage used a lower learning rate to reduce destructive drift.

Table 1: Training configuration.

Setting	Stage 1	Stage 2 continuation
Policy initialization	Base model	Stage 1 checkpoint
KL reference	Base model	Base model
Train rows	0–499	500–999
Updates	500	500
Group size	4	4
Learning rate	5×10^{-6}	2×10^{-6}
KL beta	0.05	0.05
Sampling	temperature 0.7, top-p 0.95	temperature 0.7, top-p 0.95

2 Reward Design

The reward is deterministic and rule-based. The dominant signal is final-answer correctness, with smaller shaping terms for parseable and consistently formatted outputs.

Table 2: Reward components.

Condition	Reward contribution
Final answer numerically correct	+1.0
Completion ends with a recognized final answer line	+0.2
Answer is parseable as a number or expression	+0.1
Parseable answer is wrong	-0.4
No parseable answer	-0.5
Completion exceeds character budget	-0.2
Generation reaches max length without EOS	-0.3

The intended format is a final line of the form `Answer: <number>`. The scorer also accepts common variants such as `Final Answer: <number>` and boxed LaTeX answers as fallbacks. Numeric equality handles commas, currency symbols, decimal/integer equivalence, fractions, and simple LaTeX fractions. If `math-verify` is installed, it is used as an additional correctness verifier.

3 Held-Out Results

Table 3 compares the base model, the 500-update model, and the 1000-update continuation model on the same 200 held-out GSM8K test examples.

Table 3: Held-out GSM8K test performance on 200 examples.

Model	Correct	Accuracy	Answer format	Parseable	Avg. reward
Base	133/200	66.5%	155/200, 77.5%	200/200	0.786
GRPO-500	140/200	70.0%	196/200, 98.0%	200/200	0.876
GRPO-1000	142/200	71.0%	195/200, 97.5%	200/200	0.889

Relative to the base model, GRPO-1000 improved accuracy by **+4.5 percentage points** and answer-format compliance by **+20.0 percentage points**. Truncation remained 0/200 for all runs.

4 Matched Ablations

Two single-change ablations used the same seed, train rows 0–999, two 500-update stages, sampling settings, and 200-example held-out evaluation as GRPO-1000. The no-KL variant set $\beta = 0$. The no-shaping variant removed only the positive +0.2 final-format and +0.1 parseability rewards; it retained correctness rewards and all wrong-answer, invalid-output, length, and truncation penalties.

Table 4: Matched ablations on the same 200 held-out GSM8K examples.

Run	Correct	Accuracy	Answer format	Parseable	Truncated
Full GRPO-1000	142/200	71.0%	195/200, 97.5%	200/200, 100%	0/200
No KL ($\beta = 0$)	136/200	68.0%	195/200, 97.5%	200/200, 100%	2/200
No positive shaping	138/200	69.0%	169/200, 84.5%	200/200, 100%	1/200

Neither ablation improved held-out accuracy. Removing KL reduced accuracy by 3.0 percentage points while leaving final-answer compliance unchanged, supporting the value of anchoring the policy to the base-model reference. Removing positive shaping reduced accuracy by 2.0 points and final-answer compliance by 13.0 points, while measured parseability remained saturated at 100%. This separates the effect of parseability as a metric from the reward's effect on consistently ending in the requested final-answer format.

The no-shaping average reward is not compared against the full-reward run because its reward scale is intentionally different. These are matched single-seed sensitivity checks, not definitive causal estimates.

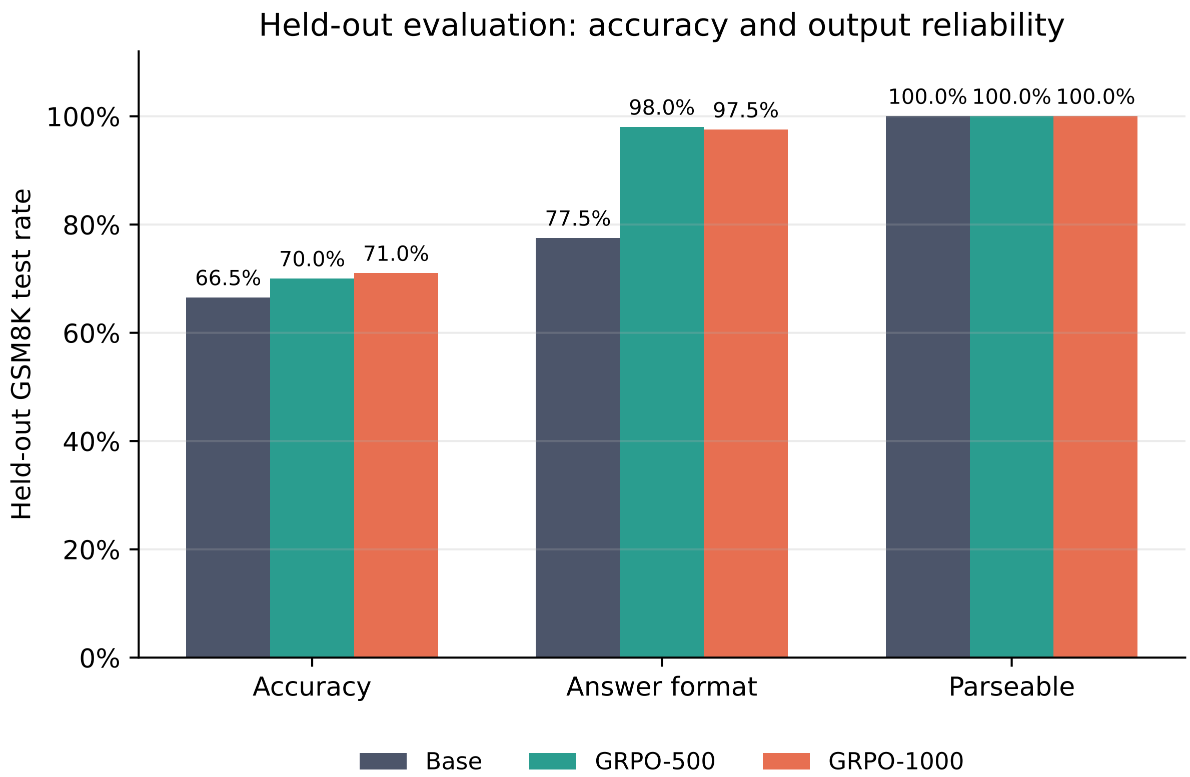

Figure 1: Accuracy, final-answer format compliance, and parseability on held-out GSM8K examples.

5 Paired Flip Analysis

Because every model was evaluated on the same 200 test examples, the most useful comparison is paired: which examples changed from wrong to correct, and which regressed?

Table 5: Paired correctness flips versus the base model.

Model	Both correct	Base wrong, trained correct	Base correct, trained wrong	Both wrong	Net gain
GRPO-500	118	22	15	45	+7
GRPO-1000	118	24	15	43	+9

For GRPO-1000, 24 examples improved from wrong to correct, while 15 regressed from correct to wrong, giving a net gain of 9 examples. This matches the aggregate gain from 133/200 to 142/200. A conservative exact paired sign test on the 39 discordant examples gives $p = 0.200$, so this should be presented as a promising pilot result rather than a statistically conclusive benchmark result.

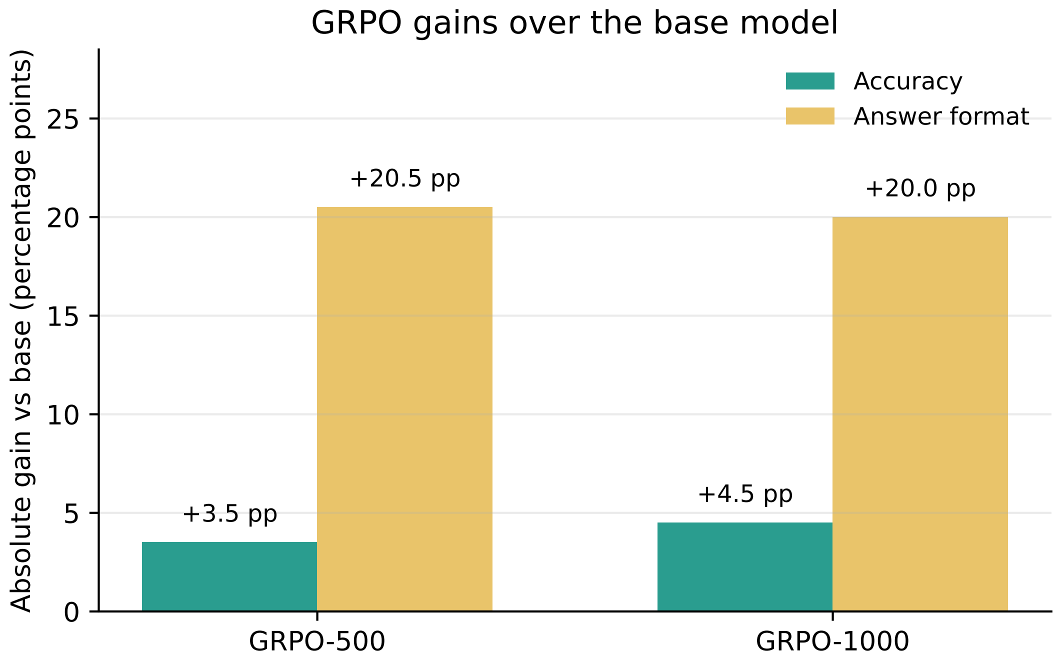

Figure 2: Absolute gains over the base model. The largest gain is in final-answer format compliance, but accuracy also improves.

6 Training Dynamics

The training logs show that the model learned the requested output format quickly and maintained stable GPU behavior. The continuation stage achieved higher train-time completion correctness, but the held-out accuracy gain from Stage 1 to Stage 2 was modest.

Table 6: Training-time completion statistics.

Run	Sampled completions	Correct completions	Format compliance	Avg. step reward	Truncated
GRPO-500	2000	1401, 70.1%	1753, 87.7%	0.856	0
Continuation	2000	1547, 77.4%	1862, 93.1%	0.969	0

7 GRPO Signal Quality

GRPO relies on relative reward within a group of completions. The most informative groups are those where some completions are correct and others are wrong. Saturated groups with 4/4 correct completions provide little math-correctness signal, though they can still reinforce formatting.

Table 7: Group-level correctness distribution.

Run	0/4	1/4	2/4	3/4	4/4	Mixed groups
GRPO-500	51	46	70	117	216	233/500
Continuation	33	34	59	101	273	194/500

The continuation run had more saturated 4/4-correct groups, which suggests the policy was already

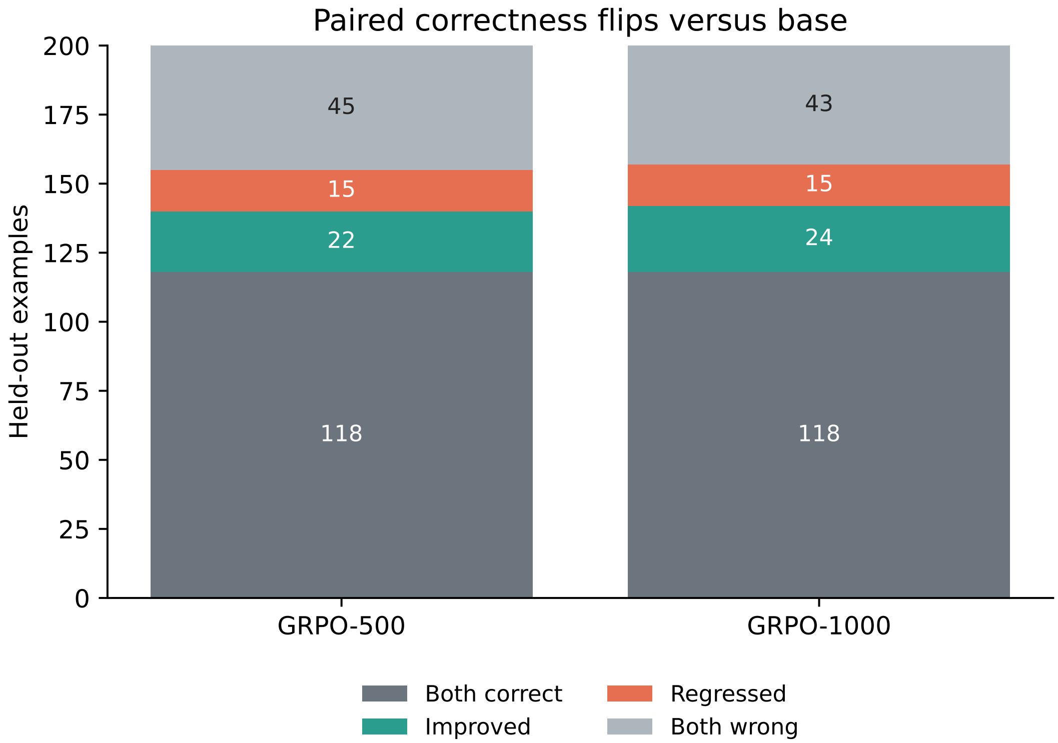

Figure 3: Paired correctness outcomes versus the base model.

strong on many sampled prompts. This is consistent with the modest additional held-out gain from 70.0% to 71.0%.

8 Resource Profile

The experiment fit comfortably on the Turing cluster's RTX 6000 Ada GPU. Peak logged memory was about 25.0 GiB in Stage 1 and 23.6 GiB in continuation, well below the 48 GiB device capacity. Peak logged temperature was 72 C.

9 Conclusion

The 1000-update TinyGRPO-Math run produced a measurable pilot improvement over the base model:

- Accuracy improved from **66.5%** to **71.0%** on 200 held-out GSM8K test examples.
- Final-answer format compliance improved from **77.5%** to **97.5%**.
- Parseability stayed at **100%**; truncation stayed at **0%**.
- The result was achieved with deterministic rule-based rewards, without training a reward model.
- Matched ablations found that neither removing KL nor removing positive format/parseability shaping improved held-out accuracy.

The strongest honest project claim is therefore: *GRPO with deterministic math rewards improved held-out GSM8K accuracy by 4.5 percentage points and final-answer compliance by 20.0 percentage*

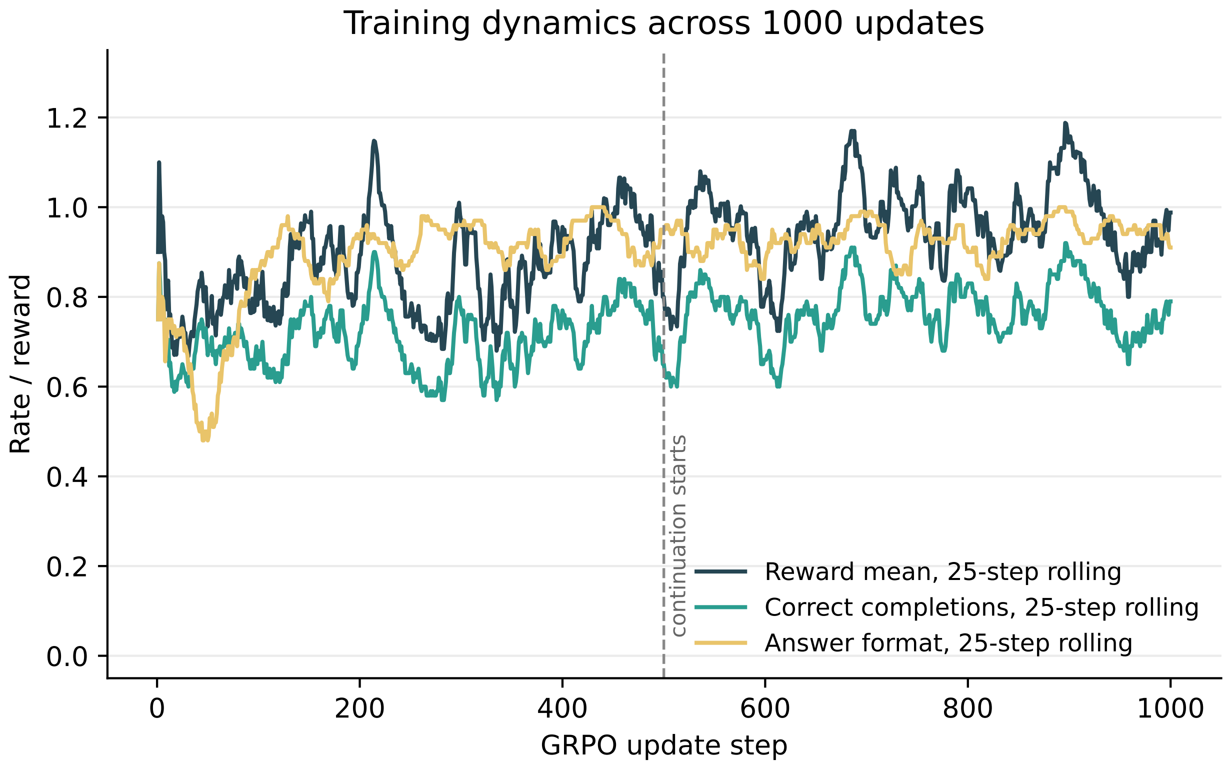

Figure 4: Rolling training curves across 1000 updates. The vertical line marks the transition from the first stage to continuation training.

points for a compact 1.5B instruction model. The matched ablations support the chosen KL reference and positive shaping terms, but are one-seed sensitivity checks rather than conclusive effect estimates. The main caveat is evaluation size: 200 examples is enough for a strong side-project demonstration, but not enough for a benchmark-quality statistical claim. A natural next step is repeated seeds, a full GSM8K test evaluation, and ablations over group size and dataset slice.

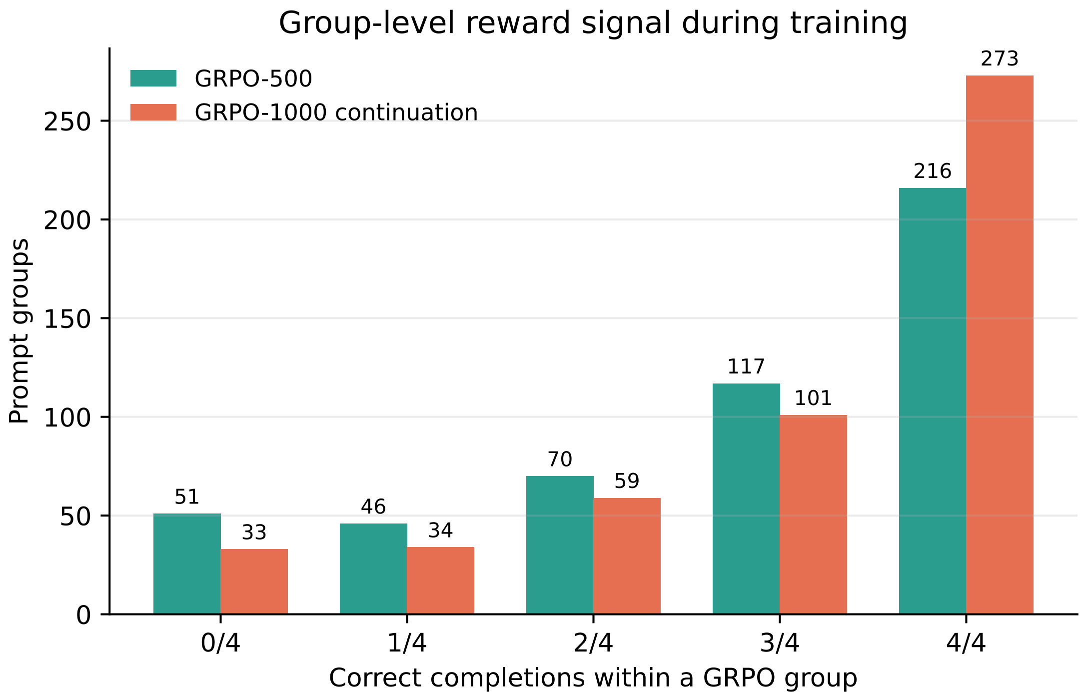

Figure 5: Correct completions per prompt group during training. Mixed groups are most useful for GRPO's relative advantage signal.

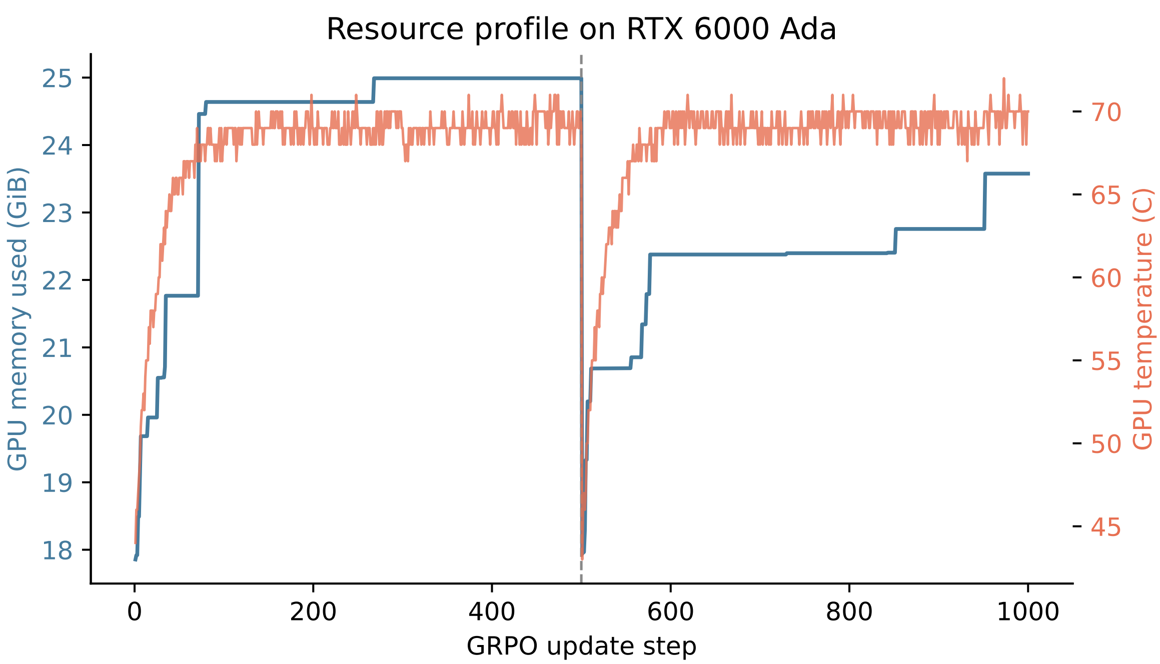

Figure 6: GPU memory and temperature over the two training stages.